Growing Hypergraphs with Node Labels

Francis Cataldo*,[1] Violet Ross*,[2] and Philip S. Chodrow[1]

[1] *Department of Computer Science, Middlebury College, Middlebury, VT, USA*

[2] *Department of Computer Science, University of Colorado at Boulder, Boulder, CO, USA*

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We study a generative mechanistic model of hypergraph growth in which nodes possess binary labels which generalizes prior work on edge-copying growing hypergraphs. Edges form in this model as noisy copies of previous edges, where the transmission of nodes from one edge to the next depends on the labels. We derive a power law for the degree distribution in this model and describe the long-term dynamics of the joint distribution of labels contained in edges. We then demonstrate that this model can be learned from data: we can both estimate the parameters of the model via (stochastic) expectation maximzation and perform community detection of the node labels via maximum-likelihood estimation.

I. INTRODUCTION

The use of a network to model a complex system is often carried out with one of two aims: modeling dynamics *on* a network or modeling the dynamics *of* a network. This work falls into the latter category — we model the growth of a complex system via the growth of its network representation. Specifically we focus on a type of complex system called a higher-order network and a type of network representation called a hypergraph.

Higher-order networks (HONs) are complex systems composed of entities and the multi-way relationships between them. Many real-world systems are composed of higher-order interactions. Social interaction (both in-person [1] and online [2]), academic co-authorship [3], and ecological systems [4] all contain interactions consisting of more than two entities.

It is often useful to represent an HON using a data structure that models its interactions. Hypergraphs, bipartite graphs, and simplicial complexes can all be used to model HONs [5]. Dyadic networks, or graphs, can be used to model dynamics on HONs when the equations of the dynamics properly account for multiway interaction [6]. However, when multiway interactions are not incorporated into the model specification, down-the-line analyses suffer [2, 7].

A hypergraph $\mathcal{H}$ is composed of a set of nodes $\mathcal{N}$ and a set of hyperedges between them $\mathcal{E}$. A hyperedge is an edge that can contain any number of nodes. We propose a model of HON growth using hypergraphs. The primary mechanism of our model is noisy copying of group interactions, or hyperedges. We therefore choose to formulate the approach as a hypergraph growth model in the interest of clear model description and analysis. Note, however, that this model could be equivalently formulated as a bipartite graph, simplicial complex, or annotated dyadic network growth model.

A. Higher order!!!

[5, 8, 9].

General hypergraph data mining:

[10]

B. Structural Properties of Higher-Order Network Data

[11]

[12]

Impact of overlap on dynamics: [13]

C. Generative Hypergraph Models

Higher-order intersections: [2]

CRU: [14]

Edge-copy model: [15]

THyME: [16]

A model based on diversity: [17]

D. Generative Models for Inference and Community Detection

Graph generative model for community detection: [18]

[19] [1, 20, 21]

Work by Yizhe Zhu, Schaub et al. on nonbacktracking methods.

[22]

Emphasize here that all of these models have the property that edges are independent conditioned on the parameters and node labels, which is arguably unrealistic for interaction data and contrasts to our setting.

E. Need to categorize

[23]

* Equal author contributions

FIG. 1. PLACEHOLDER: schematic of the model here, or alternatively a sample from the model.

F. Tooling

PyTorch: [24]
Simulated annealing? [25]
Gradient descent [26]
XGI!! [27]

II. MODEL DESCRIPTION

Our model builds on the Hyperedge Copy Model (HCM) of He et al. [15]. In their model, edges form through a combination of three distinct steps: a copy step, in which a previous edge is sampled and imperfectly copied; an extant node addition step, in which new nodes are added to the new edge from the existing node set; and a new node addition step, in which new nodes are added to the new edge from outside the existing node set. Our model replicates these mechanisms but adds label-aware parameters to regulate each.

Formally, a state of our model at time t is a hypergraph $\mathcal{H}^{(t)} = \left(\mathcal{V}^{(t)}, \mathcal{E}^{(t)}\right)$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of hyperedges. Each node $v \in \mathcal{V}$ has a binary label $z_v \in \{0,1\}$. We let $\bar{z} = 1 - z$ denote the opposite label of z. At timestep t, add edge e to $\mathcal{E}^{(t)}$ to obtain $\mathcal{E}^{(t+1)}$, where e is formed according to the following three-step process:

- **Edge-Copying**: An edge f is sampled uniformly at random from $\mathcal{E}^{(t-1)}$, and a node u is sampled uniformly at random from f. Then, each node v in f is included in e with probability η_+ if $z_v = z_u$ and with probability η_- if $z_v \neq z_u$.
- **Extant Node Addition**: We sample X_{z_u} nodes from $\mathcal{V} \setminus f$ with label z_u and add them to e, where X_z is a Poisson random variable with mean λ_+. We similarly sample $X_{\bar{z}_u}$ nodes from $\mathcal{V} \setminus f$ with label $\bar{z}_u$ and add them to e, where $X_{\bar{z}_u}$ is a Poisson random variable with mean λ_-.
- **Novel Node Addition**: We sample Y_{z_u} new nodes with label z_u and add them to e, where Y_z is a Poisson random variable with mean γ_+. We similarly sample $Y_{\bar{z}_u}$ new nodes with label $\bar{z}_u$ and add them to e, where $Y_{\bar{z}_u}$ is a Poisson random variable with mean γ_-.

In our simulations we initialize $\mathcal{H}^{(0)}$ with a single edge containing one node of each label, but any combination of hypergraph and node labels can be used as an initial condition.

A. Asymptotics

1. Joint Distribution of Labels in Edges

The asymptotics of this model can be described in terms of the joint distribution $p_{K_0,K_1}(k_0, k_1)$ of nodes of labels 0 and 1 in a uniformly random edge. This joint distribution can be obtained as the leading eigenvector of a linear map. Let $p^{(t)}(k_0, k_1)$ be the joint distribution of label counts in a uniformly random edge after t timesteps. Then, the probability $q(k_0', k_1')$ of realizing an edge with k_0' nodes of label 0 and k_1' nodes of label 1 at time $t+1$ is given by

$$q(k_0', k_1') = \sum_{k_0,k_1} T(k_0', k_1' | k_0, k_1) p^{(t)}(k_0, k_1) \ ,$$

where $T(k_0', k_1' | k_0, k_1)$ gives the probability that the edge e realized in step $t+1$ has k_0' nodes of label 0 and k_1' nodes of label 1 given that the edge f sampled to form e has k_0 nodes of label 0 and k_1 nodes of label 1. At stationarity, we must have $q(k_0', k_1') = p^{(t)}(k_0', k_1') = p^{(t+1)}(k_0', k_1')$, so the stationarity condition can be obtained by solving the linear system

$$p^{(t+1)}_{K_0,K_1}(k_0', k_1') = \sum_{k_0,k_1} T(k_0', k_1' | k_0, k_1) p^{(t)}_{K_0,K_1}(k_0, k_1) \ .$$

The stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ is therefore given by the leading eigenvector of the doubly-indexed matrix T with entries $T(k_0', k_1' | k_0, k_1)$. The entries of this matrix can be computed analytically. Conditional on choosing an edge f with k_0 nodes of label 0 and k_1 nodes of label 1, a node of label z is selected as the seed node u with probability $\frac{k_z}{k_0+k_1}$. Then, the number of nodes of labels z and $\bar{z}$ in e have distributions

$$\begin{aligned} k_z' &\sim 1 + \mathrm{Binom}\left(k_z - 1, \eta_+\right) + \mathrm{Poisson}(\lambda_+ + \gamma_+) \ , \\ k_{\bar{z}}' &\sim \mathrm{Binom}\left(k_{\bar{z}}, \eta_-\right) + \mathrm{Poisson}(\lambda_- + \gamma_-) \ . \end{aligned}$$

The probability of obtaining k_0' nodes of label 0, conditioning on k_0 and k_1, is

$$\begin{aligned} t(k_0' | k_0, k_1) = & \frac{k_0}{k_0 + k_1} \sum_{i=0}^{k_0'-1} \binom{k_0 - 1}{i} \eta_+^i (1 - \eta_+)^{k_0 - 1 - i} e^{-(\lambda_+ + \gamma_+)} \frac{(\lambda_+ + \gamma_+)^{k_0' - 1 - i}}{(k_0' - 1 - i)!} + \\ & \frac{k_1}{k_0 + k_1} \sum_{i=0}^{k_0'} \binom{k_0}{i} \eta_-^i (1 - \eta_-)^{k_0 - i} e^{-(\lambda_- + \gamma_-)} \frac{(\lambda_- + \gamma_-)^{k_0' - i}}{(k_0' - i)!} \ . \end{aligned}$$

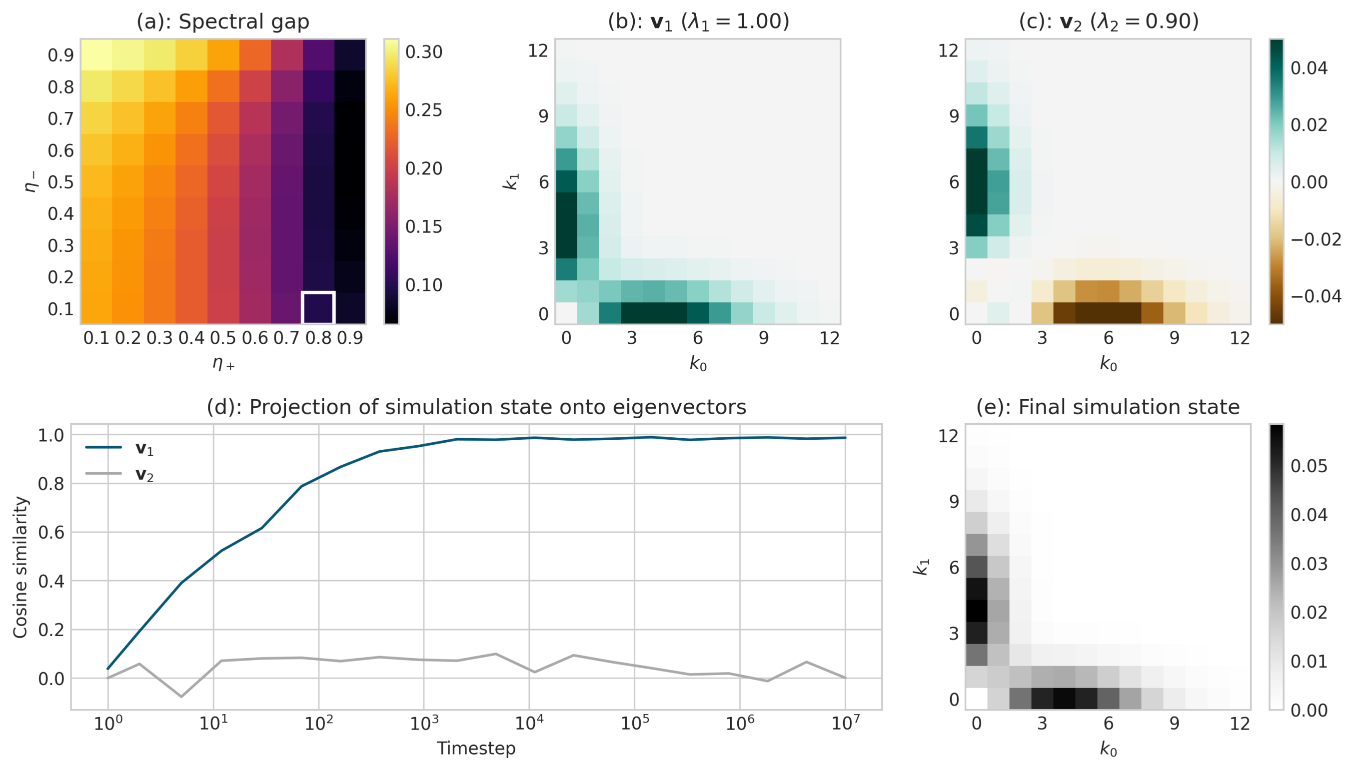

FIG. 2. Visualization of the long-run dynamics of our model, for parameters. (a): Spectral gap $1-\lambda_2$ of the linear map T as a function of η_+ and η_- for fixed parameter values $\lambda_+ = 0.5$, $\lambda_- = 0.3$, $\gamma_+ = 0.4$ and $\gamma_- = 0.2$. The highlighted values $\eta_+ = 0.8$ and $\eta_- = 0.1$ are used in the remainder of the visualization. (b): Leading eigenvector $\mathbf{v}_1$ of T, describing the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$ of label counts in edges. (c): Second eigenvector $\mathbf{v}_2$ of T. The transients of the model dynamics are dominated by the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$. (d): Euclidean projections of the simulated joint distribution $p^{(t)}_{K_0,K_1}(k_0, k_1)$ onto the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ during a simulation run with 10^7 timesteps. At each time point, we aggregate the most recent 1,000 edges to compute the joint distribution in order to approximate an "instantaneous" simulation state. (e): Empirical joint distribution $p^{(t)}_{K_0,K_1}(k_0, k_1)$ aggregated across 10^7 timesteps.

A parallel expression describes $t(k_1'|k_0, k_1)$, and the transition matrix entries are given by the product

$$T(k_0', k_1'|k_0, k_1) = t(k_0'|k_0, k_1)t(k_1'|k_0, k_1)\ .$$

In Figure 2, we show projections of the joint distribution $p^{(t)}_{K_0,K_1}(k_0, k_1)$ onto the top two eigenvectors of the matrix T during a simulation run. These eigenvectors are computed numerically. The Perron eigenvector v_1 with eigenvalue 1 corresponds to the stationary distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$. The second eigenvector v_2 with eigenvalue $\lambda_2 < 1$ corresponds to the slowest-decaying transient in the dynamics of $p^{(t)}_{K_0,K_1}(k_0, k_1)$. The spectral gap $1-\lambda_2$ is relatively small, indicating slow transients in the subspace spanned by v_1 and v_2.

2. *Degree Distribution*

The degree distribution of a hypergraph generated from our model follows an asymptotic power law with an exponent which can be computed in closed form from the model parameters and the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$. Our derivation follows that of He et al. [15], which in turn is based on a derivation by Mitzenmacher [28]. To carry out the argument, we require a mean-field assumption that, when a node v is selected from edge f in the edge-sampling step, the degree d of v is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e.

Under this assumption, it is possible to show that the degree distribution follows a power law with exponent

$$\zeta = 1 + \frac{\langle k \rangle}{1 + \eta_+(\rho_{00} + \rho_{11} - 1) + 2\eta_-\rho_{01}} \qquad (1)$$

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle .$$

In the expression for ρ_{ij} (and for $\langle k \rangle$) the expectation is taken with respect to the stationary joint distribution $p^{(\infty)}_{K_0,K_1}(k_0, k_1)$. Here, γ can be interpreted as the expected number of novel nodes added per new edge, while

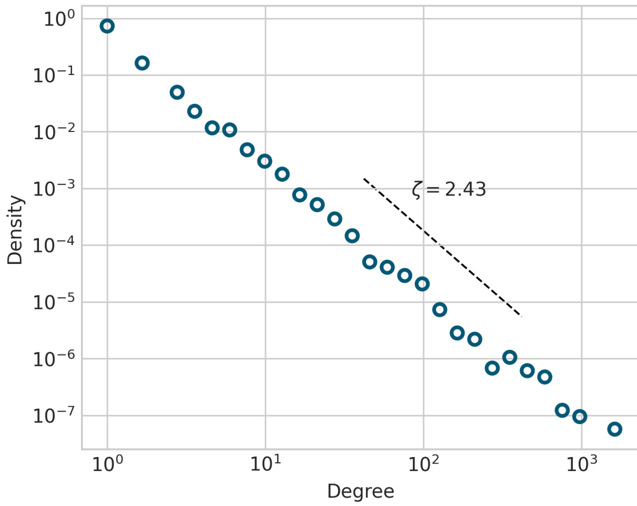

FIG. 3. Degree distribution from model simulation of 10^4 edges (points) and theoretical power law (line) with exponent given by Equation (1). The parameters in this experiment are $\eta_+ = 0.7, \eta_- = 0.3, \lambda_+ = 0.5, \lambda_- = 0.3, \gamma_+ = 0.2, \gamma_- = 0.1$.

σ can be interpreted as the expected number of nodes copied into a new edge from the sampled edge, normalized by the average degree of a node in the hypergraph.

III. MODEL INFERENCE

PC: Phil needs to establish standardized notation in this section.

A. Parameter Estimation via Stochastic EM

PC: Violet, this is your spot! Descriptions of methods/algorithms should go here, results are for later.

We introduce a Stochastic Expectation Maximization (SEM) algorithm for inferring the [INSERT MODEL NAME HERE] model parameters from a hypergraph. A traditional inference approach, maximum likelihood estimation (MLE), infers parameter values by selecting the set of parameters that are most likely to have generated the observed data. The probability that our data, the hypergraph $\mathcal{H}$, was generated using a given set of parameters θ depends not only on observable data (the hypergraph itself) but also on latent variables (the edge e' and node u selected at each timestep of the growth model). Therefore, to compute the log marginal likelihood, which MLE seeks to maximize over, we must marginalize over the latent variables, making this approach computationally expensive. Expectation Maximization (EM) algorithms also maximize over the log marginal likelihood, but avoid expensive marginalization. However, the original EM algorithm [29] computes the expected value of a sufficient statistic using all observations at each time step, which introduces another computational cost. Stochastic Expectation Maximization [30] resolves this issue by, at each timestep, computing the expectation using only one randomly selected data point.

Each iteration of the SEM algorithm consists of two steps: the E-step, in which the expected values of the sufficient statistics are calculated using the current parameter estimates, and the M-step, in which the parameter estimates are updated using the expected sufficient statistics.

Suppose we have an observed variable x and a hidden variable h. Let $f(x_i, h_i)$ be a function which computes a sufficient statistic for each of the parameters in θ and let $F_i(\hat{\theta}) = \mathbb{E}[f(x_i, h_i)|x_i]$ be the function which computes the expected sufficient statistic values given an observed x_i. Let g be a function which computes the maximum likelihood estimate of θ from the sufficient statistics. Fix a sequence of learning rates $\ell^{(1)}, \dots, \ell^{(t)}$. Each iteration, t, of SEM proceeds as follows.

E step: Sample an observation x_i from the data uniformly at random. Compute the expected sufficient statistics $F_i(\hat{\theta}^{(t-1)})$ using the randomly selected data point x_i and the parameter estimates $\hat{\theta}^{(t-1)}$ from the previous timestep. Generate an new estimate of sufficient statistics $s^{(t)}$ according to the update rule: $s^{(t)} = (1 - \ell^{(t)})s^{(t-1)} + \ell^{(t)} F_i(\hat{\theta}^{(t-1)})$.

M step: Compute the new parameter estimates $\hat{\theta}^{(t)}$ from the updated sufficient statistics: $\hat{\theta}^{(t)} = g(s^{(t)})$.

1. *Description of SEM on hypergraphs*

We use an SEM algorithm to estimate the parameters $\theta = (p, q, \gamma_{e,z_u}, \gamma_{e,\bar{z_u}}, \gamma_{e,z_n}, \gamma_{e,\bar{z_n}})$ used to generate a hypergraph $\mathcal{H}$ via the [MODEL NAME] model. At each timestep, we have an observed variable e' (a hyperedge) and two hidden variables u and e (the node and the hyperedge that generated e'). Let f be a function which computes a sufficient statistic for each of the parameters in θ. Specifically, let

$$f(e, u, e') = (t_{z_u}, w_{z_u}, t_{\bar{z}_u}, w_{\bar{z}_u}, n_{e,z_u}, n_{e,\bar{z}_u}, n_{n,z_u}, n_{n,\bar{z}_u})$$

where w_{z_u} is the number of nodes with label z_u in e, $w_{\bar{z}_u}$ is the number of nodes with the label $\bar{z}_r$ in e, t_{z_u} and $t_{\bar{z}_u}$ are the numbers of nodes with label z_u and $\bar{z}_r$, respectively, copied from e to e', n_{e,z_u} and $n_{e,\bar{z}_u}$ are the numbers of external nodes with labels z_u and $\bar{z}_r$ in e', and n_{n,z_u} and $n_{n,\bar{z}_u}$ are the numbers of new nodes with labels z_u and $\bar{z}_r$ in e'. The sufficiency of these statistics follows from standard results [?]. Then the function

$$\begin{aligned} F_{e'}(\hat{\theta}^{(t-1)}) &= \mathbb{E}[f(e,u,e')|e', \hat{\theta}^{(t-1)}] \\ &= \frac{\sum_{e\in\mathcal{E}_{\text{prev}}} \sum_{u\in e\cap e'} p(e,u,e'|\hat{\theta}^{(t-1)}) f(e,u,e')}{\sum_{e\in\mathcal{E}_{\text{prev}}} \sum_{u\in e\cap e'} p(e,u,e'|\hat{\theta}^{(t-1)})} \end{aligned}$$

computes the expected values of the sufficient statistics at time t given edge e'. Finally, the maximum likelihood estimate of θ is given by

$$g(s^{(t)}) = \left(\frac{t_u}{w_{z_u}}, \frac{t_{\bar{z}_u}}{w_{\bar{z}_u}}, n_{e,z_u}, n_{e,\bar{z}_u}, n_{n,z_u}, n_{n,\bar{z}_u} \right).$$

Each iteration of our SEM algorithm for hypergraphs then proceeds following the E- and M-steps detailed above.

B. Community Detection via MLE

PC: Frannie, this is your spot! Descriptions of methods/algorithms should go here, results are for later.

IV. INFERENCE RESULTS

How did we do on real and synthetic data?

V. CONCLUSION

Future directions:

Edge relevance-decay. Background interaction events. Work on scalability of community detection and parameter estimation. Models with multiple edge labels, e.g. planted partition (symmetric) models.

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Appendix A: Appendix

1. Power-Law Degree Distribution

Our derivation follows that of He et al. [15], which in turn is based on a derivation by Mitzenmacher [28].

At stationarity, the mean degree $\langle d \rangle$ of a node satisfies

$$\langle d \rangle = \frac{m}{n} \langle k \rangle \ , \tag{A1}$$

where $\langle k \rangle$ is the mean edge size (computable from the stationary distribution $p_{K_0,K_1}(k_1, k_2)$), m is the number of edges, and n is the number of nodes. In expectation, there are $\gamma \triangleq \gamma_- + \gamma_+$ new nodes added per new edge, which means that $\frac{m}{n} \to \frac{1}{\gamma}$ as $m, n \to \infty$ in expectation. So, we have

$$\langle d \rangle = \frac{\langle k \rangle}{\gamma} \ . \tag{A2}$$

To proceed, we make a mean-field assumption that the degree d of a node is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e. Define the moments

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle \ . \tag{A3}$$

We'll first calculate the expected number of nodes of label 0 sampled in the edge-sampling step. Suppose that an edge f is sampled with K_0 nodes of label 0 and K_1 nodes of label 1, and $K_0 + K_1 = K$ total nodes. Conditional on this event, the probability that a node of label 0 is selected as the seed node u is $\frac{K_0}{K}$, and when this occurs, the expected number of nodes of label 0 in e is $1+\eta_+(K_0-1)$. If, on the other hand, a node of label 1 is selected as the seed node u, which occurs with probability $\frac{K_1}{K}$, then the expected number of nodes of label 0 in e is $\eta_- K_0$. So, the expected number of nodes of label 0 copied into the new edge e from f is

$$\begin{aligned}\sigma_0 &\triangleq \left\langle \frac{k_0}{k}\left(1 + \eta_+(k_0 - 1)\right) + \frac{k_1}{k}\eta_- k_0 \right\rangle & \text{(A4)}\\ &= 1 + \eta_+\left[\rho_{00} + \rho_{11} - 1\right] + \eta_- \rho_{01} \ . & \text{(A5)}\end{aligned}$$

Similarly, the expected number of nodes of label 1 copied into the new edge e from f is

$$\begin{aligned}\sigma_1 &\triangleq \left\langle \frac{k_1}{k}\left(1 + \eta_+(k_1 - 1)\right) + \frac{k_0}{k}\eta_- k_1 \right\rangle & \text{(A6)}\\ &= 1 + \eta_+\left[\rho_{00} + \rho_{11} - 1\right] + \eta_- \rho_{01} \ . & \text{(A7)}\end{aligned}$$

We let

$$\sigma \triangleq \sigma_0 + \sigma_1 = 1 + \eta_+(\rho_{00} + \rho_{11} - 1) + 2\eta_- \rho_{01} \ . \tag{A8}$$

Importantly, σ depends only on the parameters and the moments of the joint distribution $p_{K_0,K_1}(k_1, k_2)$, which can be computed via linear algebra as described in Section II A 1.

We now invoke the mean-field assumption. Under this assumption, the probability that a single node selected in the sampling step has degree d is $dp_d^{(t)}$, where $p_d^{(t)}$ is the fraction of nodes with degree d at time t, since that node participates in d edges, independently of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e. The expected number of nodes of degree d sampled in the edge-sampling step is then $\sigma \frac{dp_d^{(t)}}{\langle d \rangle}$, and the expected *change* in the count of degree d nodes through the edge-sampling step is

$$\frac{\sigma}{\langle d \rangle}\left((d-1)p_{d-1}^{(t)} - dp_d^{(t)}\right) \tag{A9}$$

Similarly, the expected number of nodes which are sampled in the extant node addition step is $\lambda \triangleq \lambda_+ + \lambda_-$. Since this step does not involve any selection proportional to degree, the expected change in the count of degree d nodes through this step is

$$n^{(t+1)}p_d^{(t+1)} - n^{(t)}p_d^{(t)} = \frac{\sigma}{\langle d \rangle}\left[(d-1)p_{d-1}^{(t)} - dp_d^{(t)}\right] + \lambda(p_{d-1}^{(t)} - p_d^{(t)}) \ . \tag{A10}$$

We know that at stationarity, $n^{(t+1)} - n^{(t)}$ is in expectation equal to $\gamma \triangleq \gamma_+ + \gamma_-$, the expected number of novel nodes, while at stationarity $p_d^{(t+1)} = p_d^{(t)} = p_d$ independent of t. Thus, we obtain

$$p_d \gamma = \frac{\sigma}{\langle d \rangle}\left[(d-1)p_{d-1} - dp_d\right] + \lambda(p_{d-1} - p_d) \ .$$

Rearranging, we obtain

$$\begin{aligned}\frac{p_d}{p_{d-1}} &= \frac{\frac{\sigma}{\langle d \rangle}(d-1) + \lambda}{\frac{\sigma}{\langle d \rangle}d + \gamma + \lambda}\\ &= 1 - \frac{\frac{\sigma}{\langle d \rangle} + \gamma}{\frac{\sigma}{\langle d \rangle}d + \gamma + \lambda} \ .\end{aligned}$$

Allowing d to become large (which describes the tail of the degree distribution), we have

$$\begin{aligned}\frac{p_d}{p_{d-1}} &\approx 1 - \frac{1}{d}\frac{\frac{\sigma}{\langle d\rangle} + \gamma}{\frac{\sigma}{\langle d\rangle}} \\ &= 1 - \frac{1}{d}\left(1 + \frac{\gamma}{\sigma/\langle d\rangle}\right) \\ &= 1 - \frac{1}{d}\left(1 + \frac{\langle k\rangle}{\sigma}\right) .\end{aligned}$$

a relation which corresponds to a power law with exponent $\zeta = 1 + \frac{\langle k\rangle}{\sigma}$. Explicitly calculating this exponent requires the moments $\langle k\rangle$, ρ_{00}, ρ_{11}, and ρ_{01}, which can be computed from the stationary joint distribution $p_{K_0,K_1}(k_1, k_2)$.